

Primitive Defects and Causal Circulations

From Kovačević's positive factorization to an exact one-step decomposition of quartic zero repair

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Abstract

The motivation for this note is a very specific structural feature of Kovačević's recent sticky-channel argument. There, an entire indexed family of coefficientwise domination inequalities is compressed to one primitive defect:

$$D_m(z) = G(z)^m - \gamma G(z)^{m+d} = G(z)^{m-1} (G(z) - \gamma G(z)^{d+1}).$$

The multiplier G^{m-1} is coefficientwise nonnegative, so positivity of one primitive defect propagates to every index m .

The quartic zero-repair problem has an exact positive identity of its own, but at first it appears weaker: the recursion determines a parent only after averaging over causal descendants. The missing operation is therefore not another estimate but a positivity-compatible causal disintegration.

Re-examining the one-step recursion reveals a sharper fact. Every positive zero-repair split can be represented as a canonical comonotone split plus a nonnegative field of elementary 2×2 crossing intensities. All crossings lie in the kernel of the parent aggregation map. They preserve normalization, both marginals, and average inventory, while changing the history-by-history allocation that controls future viability. Against a test function of the inventory, one crossing is priced exactly by a discrete second difference.

For the exact depth-two optimum, the entire correction from the canonical comonotone split to a viable depth-two tree is achieved by a single crossing of intensity $1/36$. Thus the one-step causal freedom admits an exact primitive-generator description. The remaining open problem is now more precise: can the resulting crossing fields themselves be compressed across histories and scales, in the same sense that Kovačević compresses his m -indexed family to one primitive defect?

Epistemic convention. Results quoted from earlier notes are identified as such. New algebraic consequences proved in this note are marked **DERIVED HERE**. No numerical experiment is used as proof below. No historical novelty claim is attached to the elementary transportation identities; their relevance here is their role in the causal zero-repair recursion.

1 Motivation: the exact point of the analogy

The comparison with the sticky-channel paper is not based on the fact that the two channel models look alike. They do not need to.

The relevant common structure is that both problems produce a large indexed family of positivity constraints, and the central question is whether the index can be removed by identifying one primitive defect.

1.1 The sticky-channel compression

In Proposition 6 of [1], let

$$D_m(z) := G(z)^m - \gamma G(z)^{m+d}.$$

The full domination family is

$$D_m \succeq_{\text{coeff}} 0 \quad \forall m \geq 1.$$

The decisive identity is

$$\boxed{D_m = G^{m-1}D_1, \quad D_1 = G - \gamma G^{d+1}.} \quad (1)$$

Since

$$G^{m-1} \succeq_{\text{coeff}} 0,$$

one obtains

$$D_1 \succeq_{\text{coeff}} 0 \implies D_m \succeq_{\text{coeff}} 0 \quad \forall m.$$

Thus

$$\boxed{\text{indexed family} = \text{positive orbit of one primitive defect.}}$$

The same proposition goes on to characterize the domination class through the regenerative equation

$$G = (1 - \gamma)R + \gamma G^{d+1},$$

but the key structural compression already occurs in (1).

1.2 The causal question behind *Who Pays the Bill?*

Version 4 of *Who Pays the Bill?* [2] showed that, for the history-indexed two-action causal contract, dependence can be placed beyond a causal anti-diagonal so that no admissible horizon- n path can expose both sides of a coupled block. Consequently

$$C_{\infty}^{\text{causal}}(P, Q) = \max\{H(P), H(Q)\}.$$

The conceptual point is that joint latent structure can be modified in ways that remain invisible to every admissible realized path.

The later zero-repair model imposes a stricter requirement: one-step mismatches must be absorbed exactly by nonnegative history-dependent inventory laws. This turns the residual problem into a positive causal disintegration problem.

Our motivation is therefore:

$$\boxed{\begin{array}{l} \text{Kovačević removes an index by positive factorization.} \\ \text{Can the causal history index be removed in the same structural way?} \end{array}} \quad (2)$$

2 Quartic zero repair

Consider

$$P = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8}\right), \quad Q = \left(\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64}\right)$$

on $\{0, 1, 2, 3, 4\}$. Their PGFs satisfy

$$\boxed{G_P(z) - G_Q(z) = \frac{(1-z)^4}{64}.} \quad (3)$$

A depth- n positive zero-repair tree is a family of probability laws

$$\{B_h\}_{|h|\leq n}$$

on $\mathbb{Z}_{\geq 0}$ satisfying

$$Q * B_h = \sum_{x=0}^4 P(x) \delta_x * B_{hx} \quad (|h| < n). \quad (4)$$

Equivalently,

$$G_Q G_{B_h} = \sum_{x=0}^4 P(x) z^x G_{B_{hx}}. \quad (5)$$

Let $\mathcal{K}_n$ be the set of root laws admitting a positive continuation to depth n , and define

$$\mathcal{B}_n := \inf_{B \in \mathcal{K}_n} \mathbb{E}B.$$

3 Before causality, the primitive-defect factorization is already exact

The first observation is elementary but important.

Proposition 3.1 (Scalar quartic factorization). *For every $m \geq 1$,*

$$G_P^m - G_Q^m = (G_P - G_Q) \sum_{j=0}^{m-1} G_P^{m-1-j} G_Q^j. \quad (6)$$

Hence, using (3),

$$\frac{G_P^m - G_Q^m}{(1-z)^4} = \frac{1}{64} \sum_{j=0}^{m-1} G_P^{m-1-j} G_Q^j \succeq_{\text{coeff}} 0. \quad (7)$$

Proof. Equation (6) is the usual difference-of-powers identity. Every factor $G_P^{m-1-j} G_Q^j$ is a PGF and therefore has nonnegative coefficients. Substitution of (3) gives (7). $\square$

Interpretation. At the scalar convolution-power level, there is no loss of the Kovačević-type mechanism. All m -step source mismatches are generated from the single quartic defect

$$\frac{(1-z)^4}{64}$$

through a coefficientwise-positive propagator.

The difficulty appears only after lifting this scalar identity to positive history-indexed causal fibers.

4 What the causal recursion gives automatically

Iterating (5) for r generations gives

$$G_Q^r G_{B_h} = \sum_{|u|=r} P_u z^{s(u)} G_{B_{hu}}, \quad (8)$$

where

$$P_u := \prod_{i=1}^r P(u_i), \quad s(u) := \sum_{i=1}^r u_i.$$

This is an exact positive identity, but it has the form

$$\boxed{\text{parent} = \text{positive aggregate of descendants.}}$$

It does not provide a factorization of each descendant through one primitive object. Thus the original structural gap can be stated as

$$\boxed{\text{positive averaging} \neq \text{positive causal disintegration.}}$$

The rest of this note examines the one-step disintegration exactly.

5 One zero-repair split is a coupling

Fix a parent law B . Let

$$S \sim Q * B, \quad X \sim P.$$

Proposition 5.1 (Coupling representation). *A family of probability laws*

$$(B_0, \dots, B_4)$$

satisfies

$$Q * B = \sum_x P(x) \delta_x * B_x$$

if and only if

$$\boxed{\pi(x, s) := P(x) B_x(s - x)} \tag{9}$$

*defines a coupling of $X \sim P$ and $S \sim Q * B$ supported on*

$$s \geq x.$$

Proof. If the children are given, summing (9) over s gives the X -marginal P , while summing over x gives

$$\sum_x P(x) B_x(s - x) = (Q * B)(s).$$

Since B_x is supported on $\mathbb{Z}_{\geq 0}$, necessarily $s - x \geq 0$.

Conversely, from any coupling π with the stated marginals and support, define

$$B_x(j) = \frac{\pi(x, x + j)}{P(x)}.$$

The row marginal of π makes B_x a probability law, and the column marginal gives the zero-repair equation. $\square$

Thus the freedom in a positive one-step zero-repair split is exactly the freedom to choose a coupling of P and $Q * B$ inside the order wedge

$$S \geq X.$$

6 The canonical split

If such a coupling exists, then necessarily

$$Q * B \succeq_{\text{st}} P.$$

Conversely, by the one-dimensional Strassen theorem, this stochastic dominance is sufficient for a coupling with $S \geq X$.

Let

$$\pi^\uparrow$$

denote the comonotone (quantile) coupling of $X \sim P$ and $S \sim Q * B$. Because

$$Q * B \succeq_{\text{st}} P,$$

the quantile coupling also satisfies

$$S \geq X \quad \text{almost surely.}$$

For a coupling π , define the bivariate cumulative distribution

$$H_\pi(x, s) := \sum_{i \leq x} \sum_{t \leq s} \pi(i, t).$$

The comonotone coupling satisfies the Fréchet upper identity

$$\boxed{H_{\pi^\uparrow}(x, s) = \min\{F_P(x), F_{Q*B}(s)\}.} \quad (10)$$

Hence every other coupling π with the same marginals obeys

$$H_\pi(x, s) \leq H_{\pi^\uparrow}(x, s).$$

This produces nonnegative coordinates for all departures from the canonical split.

7 The crossing decomposition

For $x \in \{0, 1, 2, 3\}$ and $s \geq x + 1$, define the signed elementary plaquette

$$\boxed{\mathbb{C}_{x,s} := -\delta_{(x,s)} + \delta_{(x,s+1)} + \delta_{(x+1,s)} - \delta_{(x+1,s+1)}}. \quad (11)$$

It preserves both marginals.

Define

$$\boxed{\Gamma_{x,s} := H_{\pi^\uparrow}(x, s) - H_\pi(x, s)}. \quad (12)$$

By (10),

$$\Gamma_{x,s} \geq 0.$$

Theorem 7.1 (Canonical one-step causal circulation decomposition). *Let B be one-step viable and let π be any zero-repair coupling associated with a positive split of B . Then*

$$\boxed{\pi = \pi^\uparrow + \sum_{x=0}^3 \sum_{s \geq x+1} \Gamma_{x,s} \mathbb{C}_{x,s}, \quad \Gamma_{x,s} \geq 0.} \quad (13)$$

*For finite-support $Q * B$ the sum is finite. For finite-mean B it is well defined pointwise; the tail coefficients are summable after the finite transition region.*

Proof. The cumulative distribution of $\mathbb{C}_{x,s}$ is zero at every rectangle corner except (x, s) , where it equals -1 . Therefore the cumulative distribution of the right-hand side of (13) is

$$H_{\pi^\uparrow}(x, s) - \Gamma_{x,s} = H_\pi(x, s)$$

at every interior rectangle corner.

Both sides have the same row and column marginals. A signed measure on the discrete product lattice with zero marginal discrepancy is determined by its interior rectangle cumulatives. Hence the two measures agree.

It remains only to note why no plaquette below the wedge boundary is required. If $s \leq x$, then $S \geq X$ implies

$$\{S \leq s\} \subseteq \{X \leq x\},$$

so

$$H_\pi(x, s) = F_{Q*B}(s).$$

The same identity holds for $\pi^\uparrow$, and therefore

$$\Gamma_{x,s} = 0.$$

Thus only $s \geq x + 1$ contributes. □

Remark 7.2 (What is positive here). The plaquette $\mathbb{C}_{x,s}$ is a signed generator. The positivity statement is that every actual coupling has canonical coordinates

$$\Gamma_{x,s} \geq 0$$

relative to the comonotone coupling. Arbitrary nonnegative Γ -fields are not automatically feasible: the reconstructed coupling must still have nonnegative point masses. Thus (13) is an exact coordinate representation of the feasible coupling polytope, not yet an unconstrained positive semigroup.

This qualification is important. It is also exactly where the recursive problem remains nontrivial.

8 PGF form: crossings are invisible to the parent

Let

$$j := s - x \geq 1.$$

A crossing of intensity $\Gamma_{x,s}$ changes only the adjacent child laws B_x, B_{x+1} . In PGF form,

$$\boxed{\Delta G_{B_x}(z) = \frac{\Gamma_{x,s}}{P(x)} z^j (z - 1),} \tag{14}$$

and

$$\boxed{\Delta G_{B_{x+1}}(z) = \frac{\Gamma_{x,s}}{P(x+1)} z^{j-1} (1 - z).} \tag{15}$$

Consequently,

$$P(x) z^x \Delta G_{B_x} + P(x+1) z^{x+1} \Delta G_{B_{x+1}} = 0. \tag{16}$$

Therefore every elementary crossing lies in the kernel of the branching aggregation operator

$$A(B_0, \dots, B_4) = \sum_x P(x) z^x G_{B_x}.$$

We obtain the exact statement

$$\boxed{\text{a crossing changes the causal fibers but is completely invisible to the parent equation.}} \tag{17}$$

This is the local algebraic form of a causal circulation.

9 Conservation laws

Because a crossing preserves the X - and S -marginals, it preserves every functional depending only on either marginal.

In particular, since

$$J := S - X$$

is the child inventory,

$$\mathbb{E}J = \mathbb{E}S - \mathbb{E}X.$$

For the present quartic pair,

$$\mathbb{E}P = \mathbb{E}Q = 2,$$

so every exact split satisfies

$$\boxed{\sum_x P(x) \mathbb{E}B_x = \mathbb{E}B.} \quad (18)$$

Thus a crossing may move inventory between histories but cannot change the weighted average child inventory.

This gives a precise version of the informal accounting statement:

$$\boxed{\text{causal circulation redistributes the bill; it does not create it.}}$$

10 Dual pricing: a crossing is measured by curvature

Let $f : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}$ be any test function for which the expectations exist. Consider the weighted child functional

$$\mathcal{L}_f(B_0, \dots, B_4) := \sum_x P(x) \mathbb{E}_{B_x} f(J).$$

For the plaquette $C_{x,s}$, with $j = s - x$, the four inventory coordinates are

$$j, \quad j+1, \quad j-1, \quad j.$$

Hence:

Proposition 10.1 (Crossing-curvature identity). *One crossing of intensity $\Gamma_{x,s}$ changes $\mathcal{L}_f$ by*

$$\boxed{\Delta \mathcal{L}_f = \Gamma_{x,s} [f(j+1) + f(j-1) - 2f(j)].} \quad (19)$$

Equivalently,

$$\Delta \mathcal{L}_f = \Gamma_{x,s} \Delta^2 f(j-1),$$

where

$$\Delta^2 f(k) = f(k+2) - 2f(k+1) + f(k).$$

Proof. Pair the signed plaquette (11) with the function

$$(x, s) \mapsto f(s - x).$$

The four terms are exactly those in (19). □

Therefore every affine f is invisible to crossings:

$$\Delta^2 f = 0.$$

This immediately recovers preservation of mass and mean.

More importantly, non-affine viability or dual potentials see the crossing only through their discrete curvature.

This fits naturally with the previously derived finite-dimensional full-tree dual recursion [3]. There, normalized dual potentials satisfy the local feasibility condition

$$-\beta_h + Q^*u_h - S_x^*u_g \geq 0,$$

the raw potential has an additive constant gauge, and the causal prefix premium is an exact dual value gap with an integrated shadow-work representation.

The present calculation suggests a sharper coordinate choice:

the intrinsic local price of causal circulation is carried by discrete curvature, not by the affine part of the pot	(20)
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Equation (20) is an interpretation of the exact identity (19); it is not yet a multiscale closure theorem.

11 Every finite tree is a canonical split plus a causal circulation field

Apply Theorem 7.1 independently at every internal history.

Corollary 11.1 (History-indexed crossing coordinates). *Every finite positive zero-repair tree can be described recursively by*

- (i) *its root law $B_\emptyset$;*
- (ii) *at every internal history h , the canonical comonotone split of P and $Q * B_h$;*
- (iii) *a nonnegative crossing field*

$$\Gamma_h(x, s) \geq 0$$

describing the departure of the actual causal split from that canonical split.

Thus the history-dependent freedom has been converted from an arbitrary family of child laws into a field of local circulation coordinates.

This is a genuine reduction, but not yet the Kovačević compression: there is still one field Γ_h per history.

The remaining question is whether these fields themselves arise as the positive orbit of one primitive object.

12 Exact depth two: one crossing is enough

The previous sections are general. The quartic pair gives a particularly clean finite example.

12.1 The exact depth-two optimum

We first recall the two-coordinate viability barrier. Define

$$a_0 = 1, \quad a_{k+1} = \frac{8 + 16a_k}{27}.$$

Every

$$B \in \mathcal{K}_k$$

satisfies

$$\boxed{B(0) + \frac{7}{27}B(1) \leq a_k.} \quad (21)$$

In particular,

$$a_2 = \frac{200}{243},$$

so every $B = (b_0, b_1, \dots) \in \mathcal{K}_2$ satisfies

$$27b_0 + 7b_1 \leq \frac{200}{9}.$$

Together with

$$b_0 + b_1 \leq 1,$$

this gives

$$2b_0 + b_1 = \frac{1}{20}(27b_0 + 7b_1) + \frac{13}{20}(b_0 + b_1) \leq \frac{317}{180}.$$

Since

$$\mathbb{E}B \geq 2 - 2b_0 - b_1,$$

we obtain

$$\boxed{\mathcal{B}_2 \geq \frac{43}{180}.} \quad (22)$$

Consider now

$$\boxed{B_\emptyset = \frac{137}{180}\delta_0 + \frac{43}{180}\delta_1.} \quad (23)$$

Its mean is $43/180$, so it remains only to construct a positive depth-two continuation.

12.2 The canonical comonotone children

For $S \sim Q * B_\emptyset$, the exact law is

$$\begin{aligned} \mathbb{P}(S = 0) &= \frac{959}{11520}, & \mathbb{P}(S = 1) &= \frac{3041}{11520}, \\ \mathbb{P}(S = 2) &= \frac{223}{1152}, & \mathbb{P}(S = 3) &= \frac{317}{1152}, \\ \mathbb{P}(S = 4) &= \frac{1819}{11520}, & \mathbb{P}(S = 5) &= \frac{301}{11520}. \end{aligned}$$

The comonotone coupling with $X \sim P$ gives the canonical children

$$\begin{aligned} B_0^\uparrow &= \frac{959}{1440}\delta_0 + \frac{481}{1440}\delta_1, \\ B_1^\uparrow &= \frac{8}{9}\delta_0 + \frac{1}{9}\delta_1, \\ B_2^\uparrow &= \frac{191}{288}\delta_0 + \frac{97}{288}\delta_1, \\ B_3^\uparrow &= \frac{55}{72}\delta_0 + \frac{17}{72}\delta_1, \\ B_4^\uparrow &= \frac{1139}{1440}\delta_0 + \frac{301}{1440}\delta_1. \end{aligned}$$

This is an exact positive one-step split of the root.

However,

$$B_1^\uparrow \notin \mathcal{K}_1.$$

Indeed,

$$F_{Q*B_1^\uparrow}(1) = \frac{223}{576} > \frac{3}{8} = F_P(1).$$

The violation is exactly

$$F_P(1) - F_{Q*B_1^\uparrow}(1) = -\frac{7}{576}. \quad (24)$$

12.3 One primitive crossing repairs the tree

Apply a single plaquette

$$(x, s) = (1, 2)$$

with intensity

$$\Gamma_{1,2} = \frac{1}{36}. \quad (25)$$

The modified children are

$$\begin{aligned} B_0 &= \frac{959}{1440}\delta_0 + \frac{481}{1440}\delta_1, \\ B_1 &= \frac{8}{9}\delta_0 + \frac{1}{9}\delta_2, \\ B_2 &= \frac{223}{288}\delta_0 + \frac{65}{288}\delta_1, \\ B_3 &= \frac{55}{72}\delta_0 + \frac{17}{72}\delta_1, \\ B_4 &= \frac{1139}{1440}\delta_0 + \frac{301}{1440}\delta_1. \end{aligned} \quad (26)$$

Because the plaquette lies in the aggregation kernel, the root identity remains exact automatically.

The stochastic-dominance slacks

$$F_P(t) - F_{Q*B_x}(t), \quad t = 0, 1, 2, 3,$$

are:

x	$t = 0$	$t = 1$	$t = 2$	$t = 3$
0	$\frac{4807}{92160}$	$\frac{265}{4608}$	$\frac{913}{9216}$	$\frac{409}{4608}$
1	$\frac{1}{36}$	0	$\frac{19}{192}$	$\frac{7}{192}$
2	$\frac{743}{18432}$	$\frac{109}{4608}$	$\frac{757}{9216}$	$\frac{253}{4608}$
3	$\frac{191}{4608}$	$\frac{31}{1152}$	$\frac{193}{2304}$	$\frac{67}{1152}$
4	$\frac{3547}{92160}$	$\frac{85}{4608}$	$\frac{733}{9216}$	$\frac{229}{4608}$

All entries are nonnegative. Therefore

$$Q * B_x \succeq_{\text{st}} P \quad \forall x,$$

so every $B_x \in \mathcal{K}_1$.

We have proved:

Theorem 12.1 (Exact depth-two optimum and one-crossing realization). *For the quartic pair,*

$$\mathcal{B}_2 = \frac{43}{180}. \quad (27)$$

An optimal root is (23). Its canonical comonotone split fails future viability in exactly one child, and the single crossing

$$\Gamma_{1,2} = \frac{1}{36}$$

produces a fully viable depth-two tree.

Proof. The lower bound is (22). The construction (26) gives a depth-two tree with root mean $43/180$, proving the matching upper bound. $\square$

There is also an exact local meaning to the coefficient $1/36$. Moving crossing mass Γ at $(1, 2)$ changes the $t = 1$ slack of child 1 by

$$\frac{7}{16}\Gamma.$$

Therefore the violation (24) is cancelled precisely at

$$\frac{7}{16}\Gamma = \frac{7}{576},$$

i.e.

$$\Gamma = \frac{1}{36}.$$

Thus the active crossing is not merely a convenient rational witness: it is the exact threshold for repairing that active facet along this elementary circulation direction.

13 Audit correction: the second crossing is not needed

An earlier feasible depth-two witness contained the two crossing coefficients

$$\Gamma_{1,2} = \frac{1}{36}, \quad \Gamma_{2,3} = \frac{31}{1440}.$$

Rechecking the full one-step viability inequalities in exact rational arithmetic shows that the second crossing is unnecessary.

After applying only

$$\Gamma_{1,2} = \frac{1}{36},$$

all five children already belong to $\mathcal{K}_1$, as certified by the table above.

Hence

the exact depth-two repair can be localized to one elementary crossing plaquette.

The second crossing is another feasible motion inside the depth-two solution face; it is not required for viability.

14 Why a common anchor is more rigid than causal circulation

The common-anchor programme asks for one law

$$C_n \in \mathcal{K}_n$$

that simultaneously sits below all insured last-layer histories in stochastic order.

At depth two, a separate exact calculation shows that this fixed common floor has a positive premium. For the sharp Bernoulli reserve

$$R_* = \frac{17}{20}\delta_0 + \frac{3}{20}\delta_1,$$

every depth-two tree satisfying the common-anchor condition has

$$\mathbb{E}B_\emptyset \geq \frac{749}{2700}.$$

Since Theorem 12.1 gives

$$\mathcal{B}_2 = \frac{43}{180} = \frac{645}{2700},$$

the premium is at least

$$\boxed{\frac{749}{2700} - \frac{43}{180} = \frac{26}{675} > 0.} \quad (28)$$

The one-crossing realization explains geometrically why such a premium can appear.

A common anchor tries to move all last-layer histories toward one common positive floor. A causal crossing does something more flexible:

it gives inventory to one history and takes it from a neighboring history while keeping the parent exactly fixed.

Thus the common-anchor condition suppresses part of the kernel freedom that the causal problem is actually allowed to use.

This does not prove that common anchors are useless for upper bounds. It does show that they are not the intrinsic description of optimal causal disintegration.

15 The revised structural dictionary

The comparison with Kovačević can now be made more precisely.

	Sticky-channel tion	domina- tion	Quartic zero repair
Indexed object	D_m		causal split at history h
Canonical primitive object	$D_1 = G - \gamma G^{d+1}$		comonotone split plus elementary crossing generator
Positive coordinates	multiplier G^{m-1}		crossing intensities $\Gamma_h(x, s) \geq 0$
Exact factorization	$D_m = G^{m-1} D_1$		$\pi_h = \pi_h^\uparrow + \sum \Gamma_h \mathcal{C}$
What remains invisible	encoded in positive multiplier structure		crossings lie in $\ker A$
Scalar price	coefficientwise defect / KL slack		dual price = $\Gamma \Delta^2 f$
Full index compression	yes: all m reduce to $m = 1$		not yet: one crossing field remains per history
Remaining problem	none for Proposition 6 class		compress $\{\Gamma_h\}$ across histories/scales

The loss of analogy has therefore moved.

Initially it appeared that zero repair had only positive averaging and no useful disintegration factorization.

Theorem 7.1 shows that this is too pessimistic:

one-step causal disintegration itself has an exact primitive-generator description.

What is still missing is the *recursive* compression.

16 Relation to the dual-prefix cocycle

The earlier dual-prefix analysis [3] established on finite support truncations that the causal prefix-refinement premium is an exact dual value gap and also an integrated shadow work:

$$\Pi_{n,M} = \int_0^1 \langle \lambda_s, r_0 \rangle ds.$$

The local full-tree dual feasibility condition has the form

$$-\beta_h + Q^* u_h - S_x^* u_g \geq 0,$$

and the dual potential has the exact additive gauge

$$u \sim u + c.$$

The crossing decomposition now supplies a concrete primal coordinate system for the degrees of freedom priced by such dual potentials.

The identity

$$\Delta \mathcal{L}_f = \Gamma \Delta^2 f$$

suggests that the relevant quotient may be sharper than constants alone: affine components of a test function are invisible to every local crossing. However, only the constant gauge has previously been proved to be a free direction of the full dual problem. Therefore one must not quotient out linear modes globally without a separate theorem.

The safe conclusion is:

crossing coordinates identify the primal circulation modes; discrete curvature is their exact local dual pairing; the multiscale relation to the full dual cocycle remains open.	(29)
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17 The new thin ice

The hard problem can now be stated without reference to an arbitrary bank of child laws.

At each history h , the split is represented by

$$\Gamma_h(x, s) \geq 0.$$

These fields are locally cost-free in the sense of (18), but they determine whether the resulting child laws remain in deeper viability kernels.

Thus the central problem becomes:

Open Problem 17.1 (Recursive crossing-factorization problem). *Find an intrinsic transformation of the crossing fields*

$$\{\Gamma_h\}$$

such that depth- n viability can be generated or characterized from a bounded number of primitive positive objects.

Ideally, one seeks positive propagation operators

$$T_x$$

and one primitive crossing/curvature state Ξ such that the history-indexed circulation field satisfies a factorization of the schematic form

$\Gamma_h = T_h \Xi, \quad T_h = T_{x_t} \cdots T_{x_1}.$	(30)
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A useful weaker result would allow a uniformly bounded or summable correction to (30).

This is now the direct analogue of

$$D_m = G^{m-1} D_1.$$

The target is no longer “find a good child law at every history.” It is

find the primitive object whose positive orbit generates all required causal circulations.

18 Three possible outcomes

The revised attack has three clean outcomes.

I. Exact recursive factorization. If (30) holds on the intrinsic attainable cone, then the causal history index has been compressed in the same structural sense as the sticky-channel index m . This would be the strongest outcome.

II. Bounded-defect factorization. If

$$\Gamma_h = T_h \Xi + \varepsilon_h$$

with a correction whose total cost per dyadic scale is uniformly bounded, the exact Kovačević-style identity fails but the mechanism may still be strong enough to prove

$$\mathcal{B}_n = O(\log n).$$

III. Obstruction to recursive positive compression. A proof that no such factorization can exist would also be decisive. It would identify the precise point where the universal primitive-defect mechanism breaks under causal viability, and would force a return to a weaker exact dual criterion rather than a positive primal factorization.

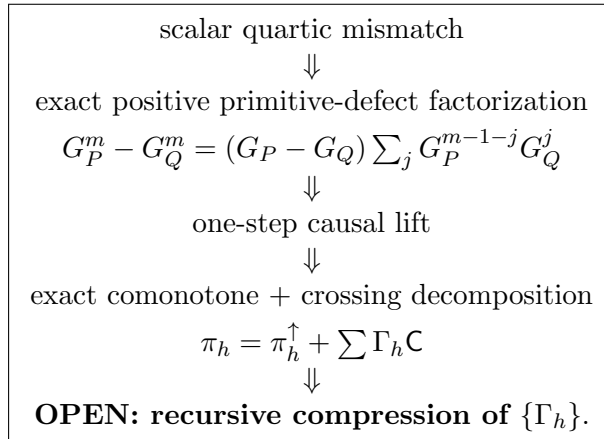
19 What has changed conceptually

Before the present calculation, the picture was

Kovačević factorizes versus zero repair only averages.

That was incomplete.

The corrected picture is



Thus two levels of the desired mechanism are already exact.

The remaining difficulty is not the existence of a local causal factorization.

It is the propagation of that factorization through future positivity and viability.

20 Question suggested by Kovačević's Proposition 6

The question we would now ask is:

In your Proposition 6, the full m -indexed domination family is generated from one primitive defect by a positivity-preserving multiplicative propagator.

For the zero-repair recursion, every one-step causal split likewise admits a canonical decomposition into a comonotone split plus nonnegative coefficients of one translated 2×2 crossing generator. These crossings are exactly invisible to the parent equation and are priced by discrete curvature in the dual variable.

Is there a natural way to iterate these crossing coordinates so that the history-indexed family of circulation fields itself becomes the positive orbit of one primitive object? Or should the failure of such a recursive factorization be expected once future viability is imposed?

This is the present form of the problem.

21 Status ledger

Statement	Status
Kovačević factorization $D_m = G^{m-1}D_1$	Published result
Quartic identity $G_P - G_Q = (1 - z)^4/64$	PROVED
Scalar convolution-power factorization (7)	DERIVED HERE
Coupling representation of one zero-repair split	DERIVED HERE
Canonical comonotone split	Standard transport fact
Crossing decomposition (13) with $\Gamma_{x,s} \geq 0$	DERIVED HERE
Crossings lie in the kernel of parent aggregation	DERIVED HERE
Crossings preserve weighted average inventory	DERIVED HERE
Crossing-curvature identity $\Delta\mathcal{L}_f = \Gamma\Delta^2 f$	DERIVED HERE
Exact depth-two optimum $\mathcal{B}_2 = 43/180$ with one-crossing realization	DERIVED HERE
Single required depth-two crossing $\Gamma_{1,2} = 1/36$	DERIVED HERE
Zero-premium common-anchor condition at $n = 2$ has positive gap $26/675$	Exact calculation
Recursive factorization of history crossing fields	OPEN
Uniform bounded surcharge / global $O(\log n)$ upper bound	OPEN

22 Conclusion

The comparison with the sticky-channel paper has become substantially more precise.

The important mechanism is not a shared notation and not merely coefficientwise positivity. It is the compression of a large indexed family through a primitive defect and positivity-compatible propagation.

For quartic zero repair, the same mechanism is now visible at two levels.

First,

$$G_P^m - G_Q^m = (G_P - G_Q) \sum_{j=0}^{m-1} G_P^{m-1-j} G_Q^j,$$

so the scalar convolution-power family is generated by one quartic defect through a positive propagator.

Second, every positive one-step causal split has the exact form

$$\pi = \pi^\uparrow + \sum_{x,s} \Gamma_{x,s} \mathbb{C}_{x,s}, \quad \Gamma_{x,s} \geq 0.$$

The entire one-step causal freedom is therefore encoded by nonnegative crossing intensities multiplying one translated local generator.

At depth two, only one such generator is needed to turn the canonical split of an optimal root into a fully viable tree.

The remaining gap is now sharply localized:

Can the history-indexed crossing fields be recursively compressed
to one primitive positive object under future viability?

(31)

That is the point at which the analogy with Kovačević is still open, and it is the point on which the next attack should concentrate.

References

- [1] M. Kovačević, *The Capacity of a Family of Sticky Channels*, arXiv:2607.28281, 2026. See especially Proposition 6 and Eq. (85).
- [2] J. Mikulík, *Who Pays the Bill? Substrate Skepticism, Stable Counterfactual Identity, Hidden Synergy, and the Causal Shannon Floor*, Version 4, 19 August 2026.
- [3] J. Mikulík, *The Dual Prefix Cocycle Attack: A finite-LP exact core and a scale-cocycle target for the quartic zero-repair upper bound*, research note, 27 August 2026.